

Financial Time Series

46-929

Homework #3

Due: Thursday, April 15, 2021, 1:30pm.

1. Download data on three-month treasury bill rates from the St. Louis Federal Reserve for 1/1/85 to 12/31/2005. You can download the series from:
research.stlouisfed.org/fred2/series/TB3MS?cid=116
 - a) Plot the data and indicate whether the data series seems to be stationary.
 - b) If the series is not apparently stationary, difference the series as few times as needed to reject the presence of a unit root using the Dickey Fuller test.
 - c) Fit an ARIMA(p,d,q) model to the series using the order_selection command, keeping d at the value you chose in the previous part. If the original series was considered to be stationary, then set $d = 0$.
 - d) Run diagnostic tests on the model. Does it seem to be a reasonable fit?
2. Consider any stock of your own choosing and any time period of at least 10 years length.
 - a) Plot the one-day log returns using the Adjusted Close data.
 - b) Using the adjusted Dickey Fuller test, assess whether the series is weakly stationary. If not, can you make it stationary by differencing?
 - c) Plot the ACF and PACF. What do you see?
 - d) Use the Arma order_select command in Statsmodels. Using the aic and bic criteria, find the suggested appropriate ARMA model(s),
 - e) From the models selected and other similar models, present the results from the best model in your judgment. Does it provide a good fit to these data? Does it pass the Ljung-Box and the adjusted Dickey Fuller tests?
3. (Ruppert and Matteson, problem 16, page 359)
Suppose you fit an AR(2) model to a time series Y_t , $1 \leq t \leq n$ and the estimates were $\hat{\mu} = 100.1$, $\hat{\phi}_1 = 0.5$ and $\hat{\phi}_2 = 0.1$. The last three observations were $Y_{n-2} = 101.0$, $Y_{n-1} = 99.5$, and $Y_n = 102.3$. What are the forecasts of Y_{n+1} , Y_{n+2} and Y_{n+3} ?
4. (Ruppert and Matteson, problem 12, page 357)
The MA(2) model $Y_t - \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$ was fit to data and the estimates are $\hat{\mu} = 45$, $\hat{\theta}_1 = 0.3$, and $\hat{\theta}_2 = -0.15$. Suppose the last two values of the observed time series and residuals are $Y_{n-1} = 39.8$, $Y_n = 42.7$, $\hat{\epsilon}_{n-1} = -4.3$ and $\hat{\epsilon}_n = 1.5$. Find the forecasts of Y_{n+1} and Y_{n+2} .